

FBPIC Baseline Simulation and Initial Analysis

Purpose

This is a baseline run before the main MPhys project. The aim is to get a working FBPIC and openPMD workflow, with detailed analysis and clear, concise figures produced. This should demonstrate the key features of laser wakefield acceleration, such as a low-density electron depletion zone with a high-density sheath. It's important to note that this is only a preliminary learning and validation run, not an optimisation project.

Computing Environment

Computing Element	Information
System	Windows laptop
Linux environment	Ubuntu through WSL
Editor	Visual Studio Code connected to Ubuntu
Python environment	Conda environment: <code>fbpic</code>
Simulation code	FBPIC
Diagnostic reader	<code>openPMD-viewer</code>
Hardware used for this run	Intel Core Ultra 5 115U CPU; integrated Intel graphics not used for CUDA acceleration

Baseline Simulation

Quantity	Baseline value
Simulation script	<code>lwfa_baseline.py</code>
Total iterations	1,800
Diagnostic interval	50 iterations
Saved outputs	36 snapshots, iterations 0-1750
Final analysed iteration	1750
Final analysed time	291.9 fs
Reconstructed E_z grid	100×800 cells

Exact Data

Parameter	Symbol	Value (exactly as written in script)	Units
Plasma electron density	n_e	4.e18*1.e6	m ⁻³
Normalised laser amplitude	a_0	4.	dimensionless
Laser waist	w_0	5.e-6	m
Pulse duration	τ	16.e-15	s
Laser wavelength	Inherent quantity of GaussianLaser	0.8e-6	m
Longitudinal resolution	N_z	800	cells
Radial resolution	N_r	50	cells
Azimuthal modes	N_m	2	N/A
Macroparticle sampling	(p_{nz}, p_{nr}, p_{nt})	(2, 2, 4)	N/A
Density-ramp length	ramp_length	40.e-6	m

Equations and theory

For an electron plasma density n_e , the plasma angular frequency is:

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \epsilon_0}}.$$

The corresponding plasma wavelength is:

$$\lambda_p = \frac{2\pi c}{\omega_p}.$$

The cold plasma wave-breaking field scale is:

$$E_0 = \frac{m_e c \omega_p}{e}.$$

These are directly calculated using the density from the analysis code rather than copying the reference values. I calculated these directly as an additional sanity check.

For electron plasma density $n_e = 4.0 \times 10^{24} \text{ m}^{-3}$:

Quantity	Calculated value
Plasma angular frequency, ω_p	$1.1283 \times 10^{14} \text{ rad s}^{-1}$
Plasma wavelength, λ_p	16.6947 m
Cold wave-breaking field scale, E_0	$192.318 \text{ GV m}^{-1}$

E_0 is a reference scale, not a strict upper limit on the field, as in this simulation it is expected there will be strongly nonlinear wakes.

Diagnostic 1: two-dimensional longitudinal field

To look for evidence of a longitudinal wake, a standard two-dimensional E_z plot was mapped. This also showed where the strongest field regions were, guiding the later density and particle diagnostics.

The longitudinal electric field was loaded with `OpenPMDTimeSeries("diags/hdf5")`, which reconstructed the $\theta = 0$ plane and converted the field from V m^{-1} to GV m^{-1} before plotting $E_z(r, z)$.

At iteration 1,750, the full reconstructed field had positive and negative extremes of:

$$E_{z,\min} \approx -693 \text{ GV m}^{-1}, \quad E_{z,\max} \approx +356 \text{ GV m}^{-1}.$$

The main wake can be clearly seen near the propagation axis. There is also a shorter-scale antisymmetric structure farther off-axis, which is expected to cancel during on-axis analysis. Subsequently, these oscillations were not treated as a direct measurement of the plasma wavelength.

As for electrons in the field, $F_z = -eE_z$, a negative E_z corresponds to a force in the positive- z direction. This assumes positive- z is the forward propagation direction.

Associated files:

- `analysis/wake/plot_ez.py`
- `figures/wake/ez_final_full_scale.png`

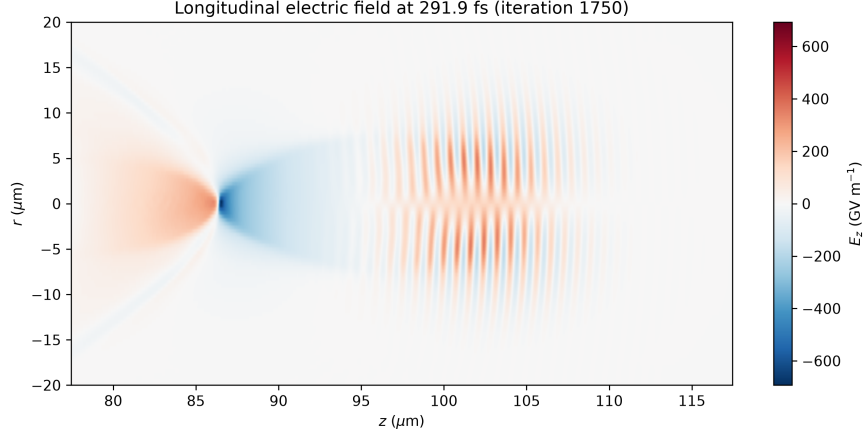


Figure 1: Two-dimensional longitudinal electric field at the final analysed iteration

Diagnostic 2: on-axis longitudinal field

The field was then reduced to an on-axis lineout in order to compare extrema, their positions and the longitudinal wake shape quantitatively.

The reconstructed slice contained the two cells immediately adjacent to $r = 0$, with their average used to estimate the on-axis field:

$$E_{z,\text{axis}}(z) = \frac{1}{2}[E_z(r_-, z) + E_z(r_+, z)].$$

The resulting lineout gave:

Measurement	Result
Maximum on-axis field	+323.86 GV m ⁻¹
Position of maximum	86.03 m
Minimum on-axis field	-679.76 GV m ⁻¹
Position of minimum	86.53 m

This showed the minimum on-axis longitudinal field to be:

$$E_{z,\text{min}} \approx -680 \text{ GV m}^{-1}.$$

This is significant when compared with the cold wave-breaking field scale, $E_0 \approx 192.3 \text{ GV m}^{-1}$, as the local minimum on-axis longitudinal field is much larger than this value.

Therefore the local minimum is approximately:

$$|E_{z,\min}| \approx 3.5E_0.$$

Importantly, the fact that the minimum on-axis field is about $3.5E_0$ does not automatically mean something is wrong, because E_0 is the characteristic cold-plasma scale, and this simulation deals with strongly non-linear wakes. However, this value alone doesn't prove that a strong non-linear wake was simulated. The feature will be investigated in further diagnostics, before taking -680 GV m^{-1} as a representative wake amplitude.

Associated files:

- `analysis/wake/plot_ez_lineout.py`
- `figures/wake/ez_final_on_axis.png`

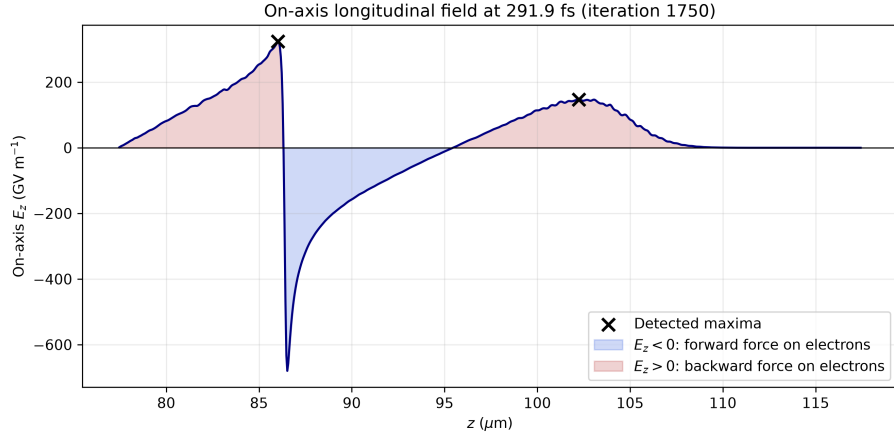


Figure 2: On-axis longitudinal electric-field lineout at the final analysed iteration

Dagnostic 3: preliminary wake-scale measurement

This diagnostic aimed to measure the separation between strong positive E_z peaks as a simple sanity check against the theoretical plasma wavelength.

Positive local maxima were detected using `scipy.signal.find_peaks`. A minimum prominence of 50 GV m^{-1} was used to reject small ripples, while the minimum allowed peak separation was decided using the expected plasma wavelength and longitudinal grid spacing.

Detected positive maxima:

Peak	Position, z	Field, E_z
1	86.025 m	323.86 GV m ⁻¹
2	102.225 m	147.40 GV m ⁻¹

The measured separation was

$$\Delta z = 102.225 \mu\text{m} - 86.025 \mu\text{m} = 16.200 \mu\text{m},$$

which is approximately 3.0% shorter than

$$\lambda_p = 16.6947 \mu\text{m}.$$

However, despite this result looking encouraging, the later E_x and laser-envelope plots placed the second maximum ($z = 102.225 \mu\text{m}$) inside the main laser region, strongly suggesting that these peaks were not a clean measurement of successive wake periods. This agreement is still a useful scale check, but to perform a proper wavelength measurement, the wake would need to be isolated behind the driver, and utilise more than one interval.

Associated files:

- `analysis/wake/measure_wake_wavelength.py`
- `figures/wake/measure_wake_wavelength.png`

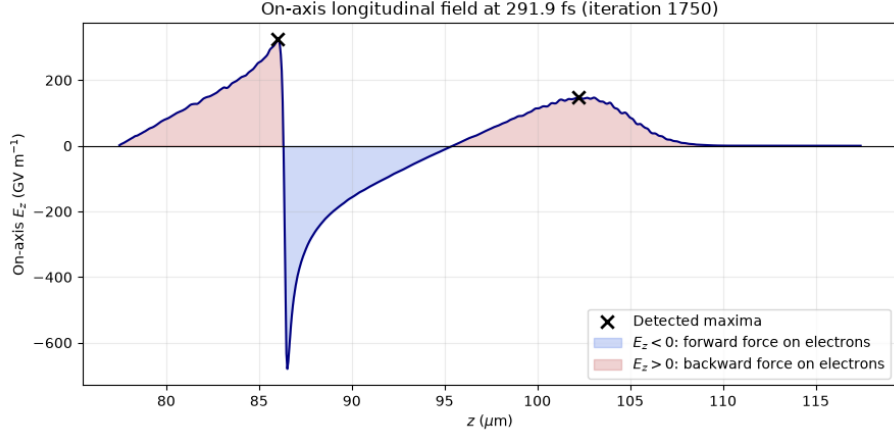


Figure 3: Preliminary comparison between detected field-peak separation and the reference plasma wavelength

Diagnostic 4.1: two-dimensional electron density

To further verify whether a wake was produced, the reconstructed electron density was next analysed to see what plasma structure sits around the large E_z feature. The charge density was loaded at $\theta = 0$ and converted to an electron-density estimate using:

$$n_e = -\frac{\rho}{e}.$$

This was then normalised to the baseline density

$$n_0 = 4.0 \times 10^{24} \text{ m}^{-3}.$$

At iteration 1,750, the reconstructed plane gave:

Measurement	Result
Minimum n_e/n_0	-0.111
Maximum n_e/n_0	127.558
Median n_e/n_0	0.999
99th percentile n_e/n_0	2.380

On this full scale plot, almost all of the colour range is taken up by a very narrow on-axis density compression close to

$$z \approx 86.5 \mu\text{m}.$$

This is the same position as the large negative E_z spike, which is consistent with a wake, however, not conclusive. The median remains almost exactly at the background density, so the $127.6n_0$ value is clearly a very local feature rather than something representative of the whole plane.

Diagnostic 4.2: scaled two-dimensional electron density

To show the ordinary wake structure more clearly, a second density plot was produced with the colour scale clipped to

$$0 \leq \frac{n_e}{n_0} \leq 5.$$

The data was left completely unchanged; only the displayed colour range is different. This makes the depleted on-axis cavity and the enhanced-density sheath much easier to see. The sheath converges near $z \approx 86.5 \mu\text{m}$, giving a structure that looks consistent with a strongly nonlinear, blowout-like wake. The next diagnostic places the laser in front of this feature.

Important diagnostic note

The small negative minimum recorded cannot represent a physical electron density. It appears in the reconstructed density around a very sharp feature, but this run alone cannot establish exactly which numerical effect produced it. Finite resolution, radial particle sampling and the limited azimuthal-mode representation are possible contributors. For the same reason, I would not treat the exact $127.6n_0$ maximum as truly converged without further checks.

Associated files:

- `analysis/wake/plot_ne.py`
- `analysis/wake/plot_density_clipped.py`
- `figures/wake/electron_density_final_full_scale.png`
- `figures/wake/electron_density_final_clipped_0_5n0.png`

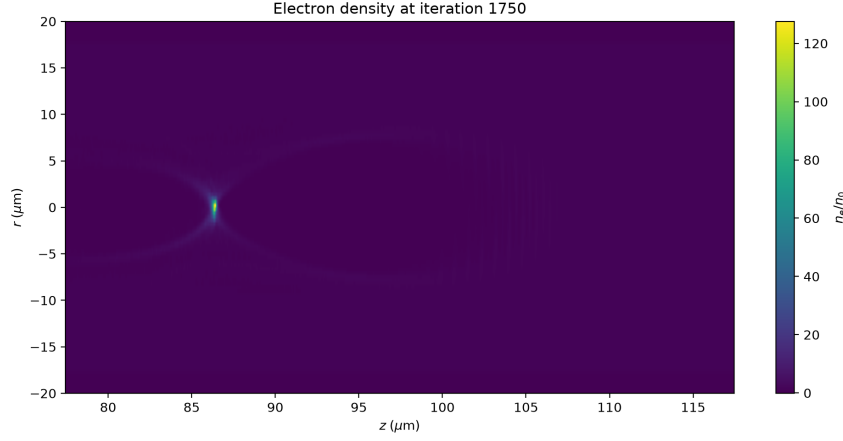


Figure 4: Electron-density reconstruction on the full colour scale

Diagnostic 5: two-dimensional transverse laser field

The electric field in x -direction, E_x , was plotted to locate the laser relative to the wake and to check that the transverse pulse and its rapid carrier oscillations were still clearly resolved at the iteration when the final snapshot was taken.

The x -component of the electric field was reconstructed at $\theta = 0$, converted from V m^{-1} to TV m^{-1} and plotted as $E_x(r, z)$. At iteration 1,750, this produced the field extrema below:

$$E_{x,\min} \approx -12.290 \text{ TV m}^{-1}, \quad E_{x,\max} \approx +12.320 \text{ TV m}^{-1}.$$

The plot showed the main laser pulse to be centred about $z \approx 103 \mu\text{m}$. This is well ahead of the strong density and E_z feature near $z \approx 86.5 \mu\text{m}$. This is

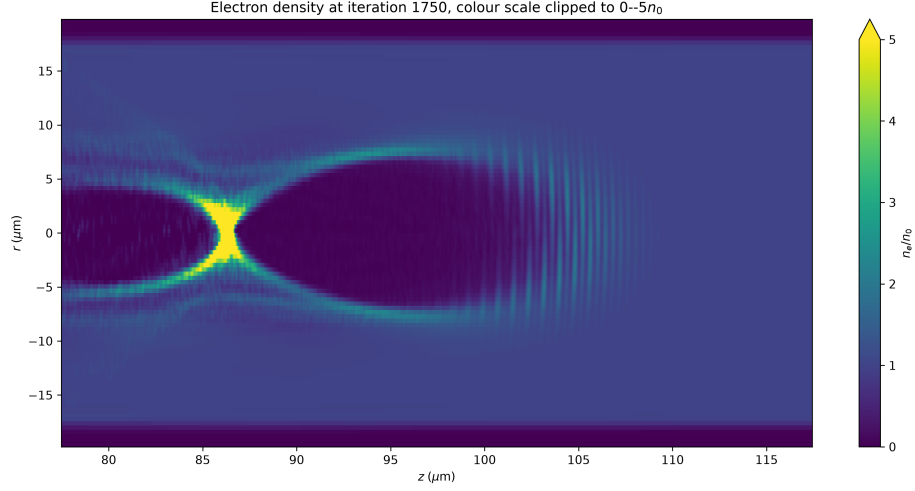


Figure 5: Electron-density reconstruction clipped to five times the background density

the expected ordering if the $z \approx 86.5 \mu\text{m}$ feature belongs to a wake behind the driving laser.

The raw E_x plot still contains the fast optical carrier, so I did not use a single field maximum as the laser position. I extracted the envelope later and used that to track the pulse more consistently.

Associated files:

- `analysis/laser/plot_ex.py`
- `figures/laser/ex_final_full_scale.png`

Diagnostic 6: longitudinal electron phase space

Having identified the wake structure and the driving laser, the next question was whether a distinct high-momentum electron population had developed. I therefore plotted longitudinal position against u_z .

The saved electron positions and normalised longitudinal momenta were loaded using `var_list=["z", "uz"]`, where

$$u_z = \frac{p_z}{m_e c}.$$

At iteration 1,750, the saved-particle distribution gave:

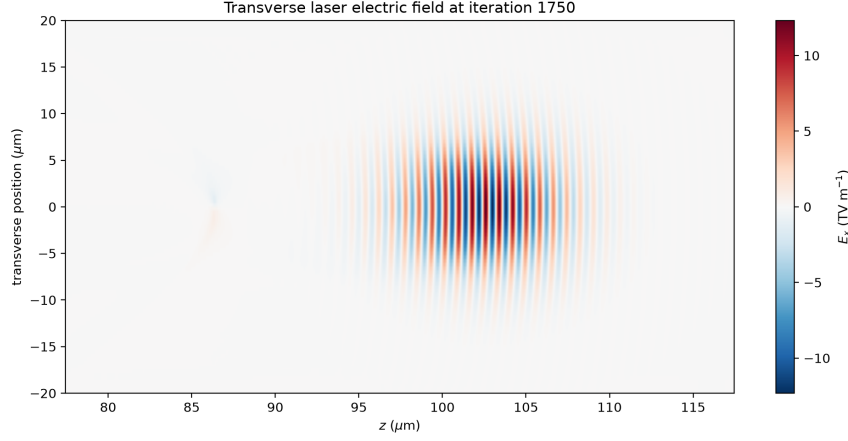


Figure 6: Two-dimensional transverse laser field at the final analysed iteration

Measurement	Result
Saved macroparticles	50,357
Longitudinal range	72.650-106.081 m
Minimum u_z	1.000
Maximum u_z	11.905
Median u_z	1.586
95th percentile u_z	4.066
99th percentile u_z	8.189

This produced two visibly different structures: a high-momentum branch sitting around $z \approx 84\text{-}87\ \mu\text{m}$ and reaching $u_z \approx 11.9$, and a more regular lower-momentum oscillatory pattern which appeared around $z \approx 101\text{-}106\ \mu\text{m}$. This further supports previous conclusions, with the E_x plot placing the laser in the second region, suggesting the oscillatory population is laser-associated, whereas the high-momentum branch is consistent with a wake.

Diagnostic note

One limitation created by the saved particle data was the diagnostic only keeping electrons with `select={"uz": [1., None]}`. Therefore, anything below $u_z = 1$ is missing from the plot, so the plot is not fully representative of the complete plasma distribution. The scatter density is also only macroparticle density, with the points not weighted by the physical number of electrons they represent.

Associated files:

- `analysis/electrons/plot_z_uz.py`

- `figures/electrons/z_uz_phase_space_final.png`

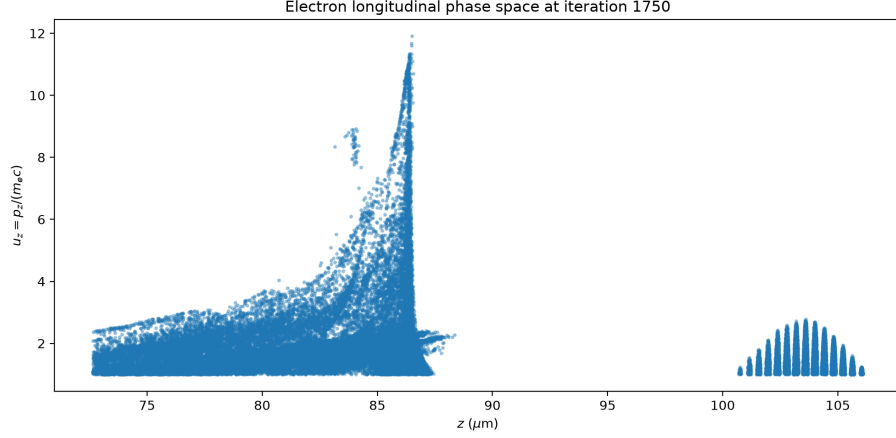


Figure 7: Final longitudinal electron phase space for the saved population

Diagnostic 7: weighted electron energy spectrum of the saved population

Logically, the next step was to convert the saved particle momenta into kinetic energy and include the macroparticle weights so that the spectrum represented physical electron charge rather than just computational particle counts.

The Lorentz factor and kinetic energy were calculated from the three normalised momentum components:

$$\gamma = \sqrt{1 + u_x^2 + u_y^2 + u_z^2}, \quad E_k = (\gamma - 1)m_e c^2.$$

At iteration 1,750, the weighted distribution gave:

Measurement	Result
Saved macroparticles	50,357
Minimum kinetic energy	0.214 MeV
Maximum kinetic energy	5.611 MeV
Weighted mean energy	1.122 MeV
Weighted RMS spread	0.965 MeV
Charge of saved $u_z \geq 1$ population	346.916 pC

The spectrum produced was broad and mostly weighted towards lower energies, with a tail reaching to about 5.6 MeV. Importantly, this cannot be called a

true “beam spectrum”, as the $u_z \geq 1$ diagnostic mixes several different forward-moving populations together.

Macroparticle weights were included through

$$Q = e \sum_i w_i,$$

which gave

$$Q_{\text{saved}} = 346.916 \text{ pC}.$$

The charge Q_{saved} represents the charge of every saved electron above the diagnostic cutoff, whether or not that electron belongs to the late wake-associated population.

Associated files:

- `analysis/electrons/plot_energy_spectrum.py`
- `figures/electrons/electron_energy_spectrum_final.png`

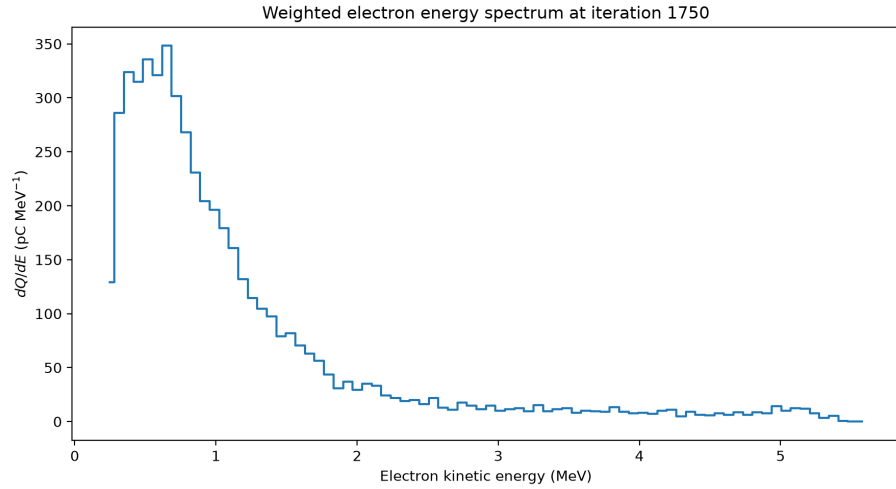


Figure 8: Weighted energy spectrum of the complete saved electron population

Diagnostic 8: maximum saved-electron energy versus time

The next key diagnostic was to calculate the maximum energy over all 36 saved snapshots to see both when high-energy particles first appeared, and whether the final 5.611 MeV value developed smoothly.

The methodology was to repeat the kinetic energy calculation from Diagnostic 7 for all 36 saved iterations. This didn’t produce any particles satisfying the

$u_z \geq 1$ diagnostic selection for the first five saved outputs, with the first saved $u_z \geq 1$ particles appearing at approximately 41.7 fs.

When kinetic energy was calculated for each iteration, the maximum saved-electron energy then showed a strongly non-monotonic evolution:

Time	Maximum kinetic energy
41.70 fs	1.139 MeV
83.39 fs	4.671 MeV
150.10 fs	2.380 MeV
200.14 fs	1.913 MeV
250.17 fs	4.287 MeV
291.87 fs	5.611 MeV

The above graph demonstrates the maximum rising quickly, then falling throughout the middle of the run and then rising again after about 200 fs. This does not mean one electron accelerated, lost energy and then accelerated again. Any macroparticle can be the maximum at each snapshot point.

The later phase-space and laser-envelope plots make this much clearer, with the early high-energy feature overlapping with the laser region, while the final high-momentum population is clearly behind the driving laser. This is a good example of why $E_{\max}$ on its own is a poor optimisation metric.

Associated files:

- `analysis/electrons/plot_max_energy_vs_time.py`
- `figures/electrons/max_electron_energy_vs_time.png`

Dagnostic 9: longitudinal-field extrema versus time

Following the unexpected maximum energy result, $E_{\max}$ was compared with the on-axis E_z extrema over the same saved iterations. This tested whether the changes in maximum particle energy followed the changes in the strongest longitudinal field.

The on-axis E_z lineout was reconstructed at every saved iteration with the positive and negative extrema recorded. The longitudinal wake grew strongly through the simulation, reaching the final values found in the earlier diagnostic which only examined iteration 1750 of:

$$E_{z,\min} \approx -680 \text{ GV m}^{-1}, \quad E_{z,\max} \approx +324 \text{ GV m}^{-1}.$$

Interestingly, between 80 and 200 fs, the negative E_z extremum becomes much stronger while the maximum saved-electron energy falls. This shows why the largest field at one position cannot by itself predict the largest particle energy.

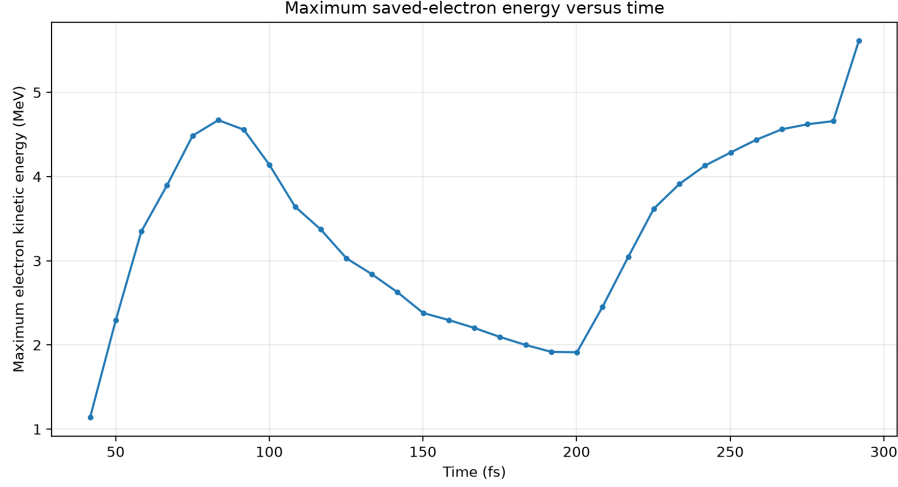


Figure 9: Maximum saved-electron energy through the baseline run

An electron's energy depends on the fields encountered along its path, and the electron giving the maximum can change between snapshots. The timing does not identify how the early particles gained energy, but the phase-space and laser-position diagnostics place that early high-energy structure with the laser rather than with the late wake-associated branch. Spatial overlap does not establish a specific acceleration mechanism.

After about 200 fs, the on-axis E_z extremum keeps growing and a separate high-momentum population develops behind the laser. The maximum energy rises again at the same time. The phase-space location strengthens the late wake-associated interpretation; the two rising curves alone would not establish that the field extremum accelerated those particular electrons.

Associated files:

- `analysis/wake/plot_ez_extrema_vs_time.py`
- `figures/wake/ez_extrema_vs_time.png`

Dagnostic 10: longitudinal phase-space evolution

As a further test, the phase space was plotted at five times to see where the high-momentum structures appeared as the run progressed.

The z - u_z distribution was compared at five representative saved iterations:

Iteration	Time	Saved macroparticles	Maximum u_z
500	83.39 fs	28,172	9.053

Iteration	Time	Saved macroparticles	Maximum u_z
900	150.10 fs	35,352	4.032
1200	200.14 fs	38,775	4.272
1500	250.17 fs	46,384	8.506
1750	291.87 fs	50,357	11.905

At 83.4 fs the largest u_z value is part of a regular sequence of peaks near the front of the saved distribution. The high-momentum feature mostly disappears by 150-200 fs, but by 250 fs, a different branch has appeared well behind the regular forward oscillations, and this later branch becomes much stronger by the final snapshot.

This is key: the phase-space structure changes, so the rise-fall-rise in $E_{\max}$ should not be interpreted as the energy history of one bunch. The snapshots do not identify individual electrons. The next diagnostic tests whether the regular forward pattern moves with the laser.

Associated files:

- `analysis/electrons/plot_z_uz_evolution.py`
- `figures/electrons/z_uz_phase_space_evolution.png`

Diagnostic 11.1: laser position and envelope evolution

To test whether the regular forward phase-space pattern is tied to the laser, I tracked the laser envelope at the same times shown in the phase-space plots.

For each selected iteration, the two cells nearest the axis were averaged to obtain an on-axis $E_x(z)$ lineout. The rapidly oscillating carrier was converted to an envelope estimate using the magnitude of the analytic signal from a Hilbert transform. The position of the largest envelope value was used as the laser-envelope peak position.

Iteration	Time	Laser-envelope peak z	Peak envelope
500	83.39 fs	39.931 m	15.509 TV m ⁻¹
900	150.10 fs	59.831 m	14.559 TV m ⁻¹
1200	200.14 fs	75.081 m	13.490 TV m ⁻¹
1500	250.17 fs	90.332 m	12.850 TV m ⁻¹
1750	291.87 fs	102.832 m	12.512 TV m ⁻¹

This shows a clear result: the laser-envelope peak moves with the regular peak-dip-peak pattern in the z - u_z plots. This is strong evidence to treat that forward oscillatory structure as laser-associated rather than part of the late wake bunch.

The peak envelope also falls from about 15.5 to 12.5 TV m⁻¹, around a 19% reduction. That shows the pulse is changing, however, this is not necessarily

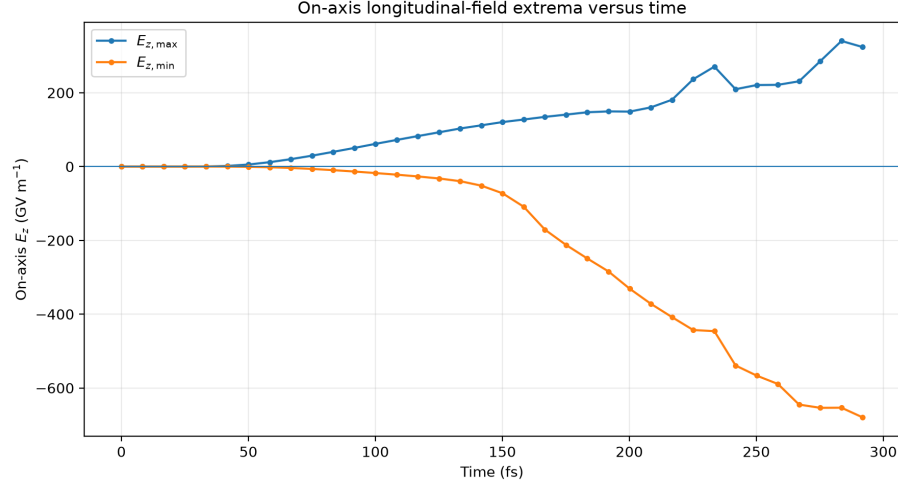


Figure 10: Evolution of the on-axis longitudinal-field extrema

a 19% laser-energy loss. This is because a change in pulse width, focusing or shape can also change the peak field, so the evidence is insufficient.

At the final snapshot the laser-envelope maximum is at

$$z_{\text{laser}} = 102.832 \mu\text{m},$$

whereas the strong high-momentum trailing branch is around

$$z \approx 86.5 \mu\text{m}.$$

This is a separation of approximately $16.3 \mu\text{m}$, close to the reference $\lambda_p = 16.6947 \mu\text{m}$. This puts the late high-momentum population at roughly the expected wake scale behind the driving laser. This is a geometrical consistency check, not a precision measurement of the nonlinear wake period.

Associated files:

- `analysis/laser/plot_laser_position_evolution.py`
- `figures/laser/laser_position_evolution.png`

Dagnostic 11.2: laser spot-size and peak- a_0 evolution

Dagnostic 11.1 showed that the peak transverse laser field fell throughout the simulation. However, the on-axis field alone could not show whether this resulted from transverse expansion, pump depletion or a change in the pulse

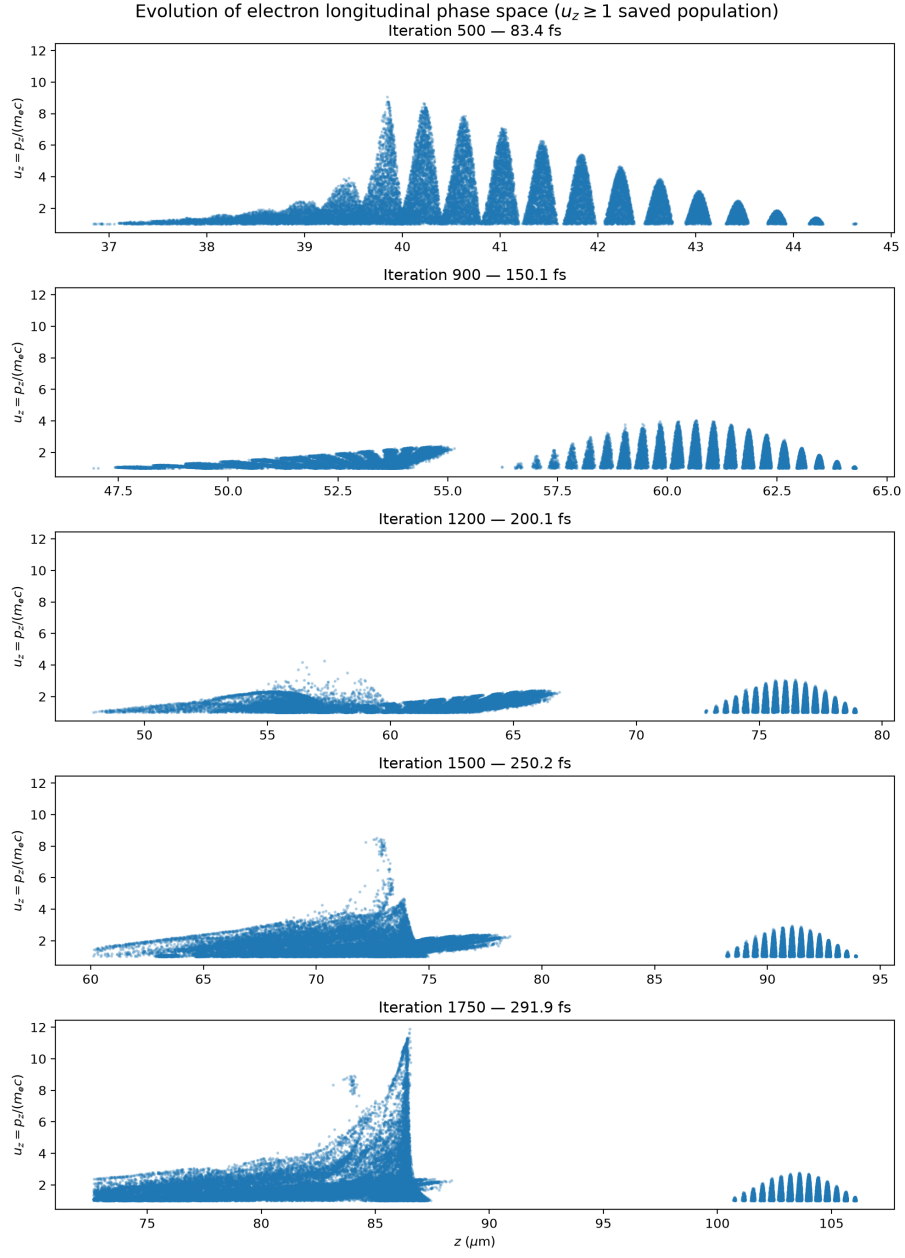


Figure 11: Evolution of the saved electron longitudinal phase space

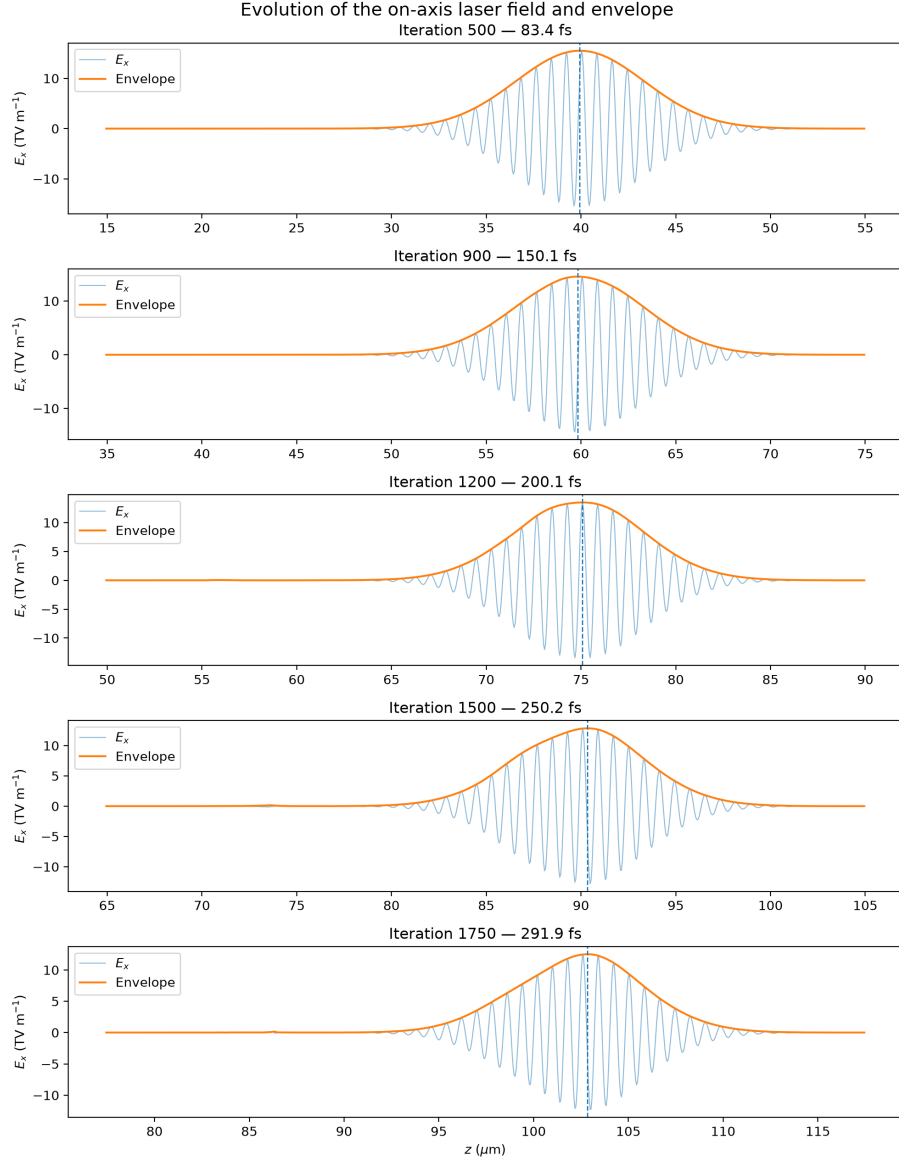


Figure 12: Evolution of the on-axis transverse laser field and its Hilbert envelope

shape. I therefore extended the analysis to measure the laser spot size and peak normalised vector potential at every saved iteration.

Method

The reconstructed $E_x(x, z)$ field contains both the rapidly oscillating laser and slower plasma-field structure. A longitudinal Fourier transform was therefore used to retain wavenumbers around the laser carrier:

$$0.5k_0 \leq k \leq 1.5k_0, \quad k_0 = \frac{2\pi}{\lambda_0}.$$

After transforming back into real space, the magnitude of the analytic signal was calculated along z using a Hilbert transform. This produced an estimate of the laser-field envelope:

$$\mathcal{E}(x, z) = |E_{x,\text{laser}}(x, z) + i\mathcal{H}[E_{x,\text{laser}}(x, z)]|.$$

The squared envelope was integrated across the longitudinal pulse window to produce a transverse weighting:

$$F(x) = \int_{\text{pulse}} \mathcal{E}^2(x, z) dz.$$

The transverse centroid and variance were then calculated from

$$\bar{x} = \frac{\int x F(x) dx}{\int F(x) dx},$$

and

$$\sigma_x^2 = \frac{\int (x - \bar{x})^2 F(x) dx}{\int F(x) dx}.$$

For a Gaussian intensity profile of the form

$$I(x) \propto \exp\left(-\frac{2x^2}{w^2}\right),$$

the spot size is related to the transverse variance by

$$w = 2\sigma_x.$$

The two cells nearest the propagation axis were averaged before taking the peak envelope value. This was converted into the normalised vector potential using

$$a_0 = \frac{eE_{\text{env}}}{m_e c \omega_0}, \quad \omega_0 = \frac{2\pi c}{\lambda_0}.$$

Vacuum and matched-waist comparisons

For comparison, the vacuum evolution of an initially focused Gaussian beam was calculated from

$$w_{\text{vac}}(z) = w_0 \sqrt{1 + \left(\frac{\Delta z}{z_R}\right)^2},$$

where

$$z_R = \frac{\pi w_0^2}{\lambda_0}.$$

For $w_0 = 5, \mu\text{m}$ and $\lambda_0 = 0.8, \mu\text{m}$, this gives

$$z_R \approx 98.2 \mu\text{m}.$$

The approximate blowout matching condition was also included:

$$k_p w_m \approx 2\sqrt{a_0}.$$

Using the plateau density and input $a_0 = 4$ gives

$$w_m \approx 10.6 \mu\text{m}.$$

This line is an approximate full-density equilibrium scale rather than a prediction that the pulse should immediately expand to $10.6, \mu\text{m}$. The matching condition also changes through the density ramp because k_p depends on the local plasma density.

Results

Measurement	Initial value	Final value	Change
Spot size, w	5.052 m	6.542 m	+29.5%
Peak a_0	3.988	3.118	-21.8%
Product $a_0 w$	20.15 m	20.40 m	+1.2%

The initial values are close to the simulation inputs of

$$w_0 = 5 \mu\text{m}, \quad a_0 = 4,$$

which provides a useful check that the diagnostic is recovering the expected laser scale.

The measured spot expands smoothly throughout the run. At the final output, vacuum diffraction predicts

$$w_{\text{vac}} \approx 6.710 \mu\text{m},$$

compared with the measured value

$$w_{\text{measured}} \approx 6.542 \mu\text{m}.$$

The measured spot is therefore only about 2.5% smaller than the vacuum prediction. This is consistent with propagation remaining largely diffraction-like over the simulated distance. The difference might reflect some plasma focusing, but without a vacuum control run or an uncertainty estimate for the measured spot size, it cannot establish that focusing occurred.

The pulse reached the beginning of the plasma at approximately 50.1 fs and the full-density plateau at approximately 183.9 fs. It travelled through only 32.8 μm of plateau plasma before the final saved output. This is approximately one-third of the vacuum Rayleigh length, so the absence of a complete spot-size oscillation is not surprising.

The decrease in peak a_0 occurs at the same time as the transverse expansion. In particular, the product $a_0 w$ changes by only about 1.2%. For a pulse that remains approximately Gaussian without a large change in duration, this is the behaviour expected when the on-axis field falls mainly because the pulse spreads transversely.

This resolves part of the ambiguity identified in Diagnostic 11. The decrease in the laser-envelope peak does not require a comparable loss of total laser energy and is largely explained by the increasing spot size. However, this analysis does not directly measure total pulse energy, so it cannot rule out some pump depletion or longitudinal pulse reshaping.

The result also provides context for the wake and particle diagnostics. The density cavity, large longitudinal field and late high-momentum population formed while the laser was expanding and its peak a_0 was falling, rather than behind a demonstrably stable matched driver. The candidate-bunch results should therefore be understood as properties of this short, transient baseline rather than an optimised accelerator configuration.

Associated files:

- analysis/laser/analyse_laser_propagation.py
- results/laser_propagation.csv
- figures/laser/laser_propagation_evolution.png

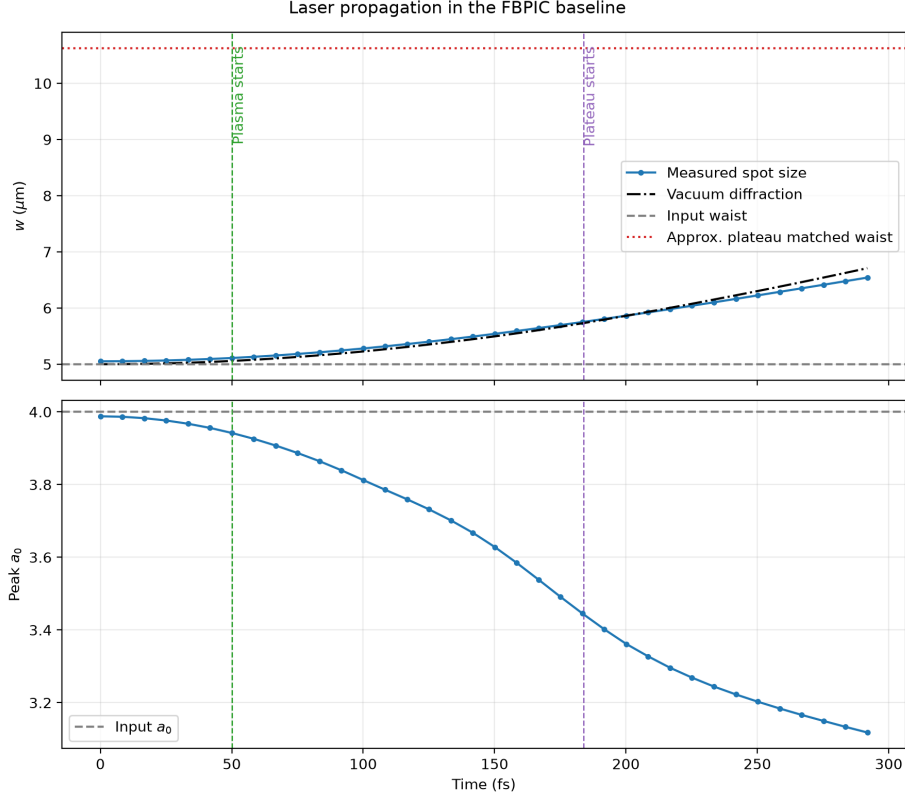


Figure 13: Evolution of the measured laser spot size and peak normalised vector potential

Diagnostic 12: robustness of the large E_z spike

Before calculating candidate-bunch properties, I returned to the sharp E_z feature identified in the earlier field and density diagnostics. The raw on-axis lineout was compared with three-cell and five-cell moving averages around the -680 GV m^{-1} minimum. A first-percentile value was also calculated within the selected wake region to provide a measure that is not controlled by the single most extreme cell. This checks whether the feature persists under modest averaging.

While the exact minimum becomes less negative after spatial averaging, the feature remains of order several hundred GV m^{-1} and the three- and five-cell averaged curves continue to reach below approximately -600 GV m^{-1} . The

first percentile of the wake-region field is also strongly negative, at roughly -580 GV m^{-1} .

The feature survives both averages, so the strong negative field is not confined to one cell. The raw -680 GV m^{-1} value is still a very local extreme, though, and should be treated as a peak rather than as the typical wake amplitude. Averaging several cells does not establish numerical convergence or exclude an artefact extending across several cells. That requires a separate comparison with different resolutions and particle sampling.

Associated files:

- `analysis/validation/test_ez_spike_robustness.py`
- `figures/validation/ez_spike_robustness.png`

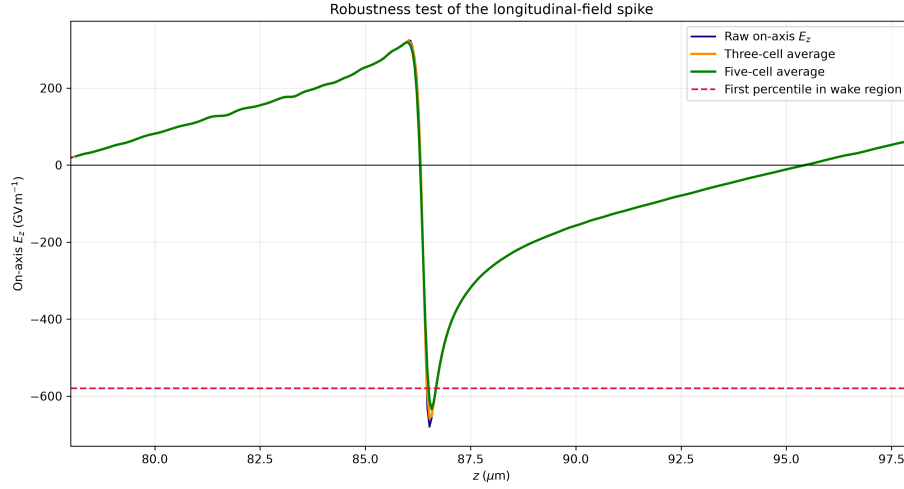


Figure 14: Longitudinal averaging and percentile test of the field spike

Diagnostic 13.1: candidate wake-bunch selection and beam metrics

With the laser and trailing high-momentum branch identified, I separated that branch from the full saved $u_z \geq 1$ population. This allowed candidate-bunch quantities to be calculated without treating every forward-moving electron as part of the same bunch.

A laser-relative coordinate was defined as

$$\xi = z - z_{\text{laser}},$$

where $\xi = 0$ corresponds to the final laser-envelope peak at $z_{\text{laser}} = 102.832 \mu\text{m}$. Negative ξ is therefore behind the laser.

Electron-bunch charge is represented as a positive charge magnitude and the candidate electron bunch was selected using

$$-20 \leq \xi \leq -14 \mu\text{m},$$

and

$$u_z \geq 4.$$

The spatial window was chosen to cover the late branch behind the laser, while $u_z \geq 4$ cuts away most of the large low-momentum population underneath it. As these are analysis choices and not unique physical boundaries, a check was performed to see how much the result moved when the cuts were changed.

For the nominal selection:

Measurement	Result
Saved macroparticles in full $u_z \geq 1$ diagnostic	50,357
Candidate-bunch macroparticles	2,609
Q_{saved}	346.916 pC
Q_{bunch}	43.254 pC
Fraction of saved charge	12.47%
Weighted mean bunch energy	3.235 MeV
Histogram peak energy	2.542 MeV
Maximum bunch energy	5.611 MeV
Weighted RMS energy spread	1.072 MeV
Relative RMS energy spread	33.14%

The selected population is clearly broad rather than quasi-monoenergetic. The 5.611 MeV value is only the end of the high-energy tail, so I would not describe this as a 5.6 MeV beam. The weighted mean, 3.235 MeV, is a better description of the selected population, and the 33% relative RMS spread shows how broad it still is.

The low-energy edge of the selected spectrum is partly imposed by the $u_z \geq 4$ momentum cut itself. The spectrum should therefore be interpreted together with the phase-space selection rather than as an independent natural low-energy boundary.

Associated files:

- `analysis/electrons/analyse_final_bunch.py`
- `figures/electrons/final_bunch_selection.png`
- `figures/electrons/final_bunch_energy_spectrum.png`

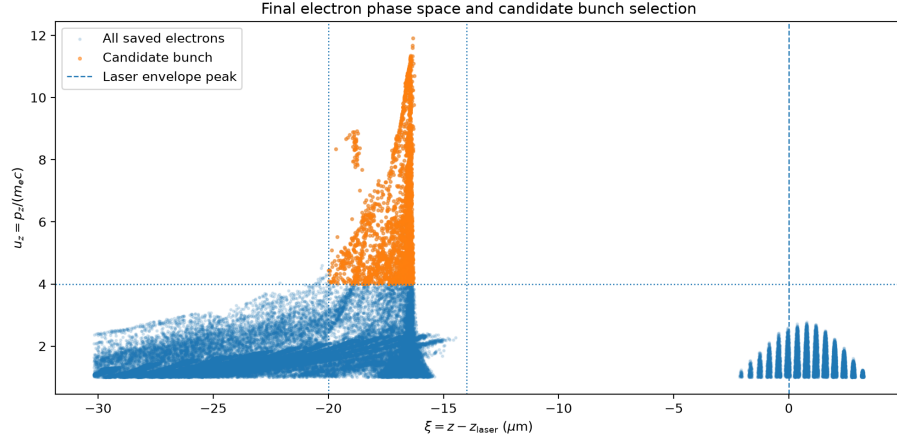


Figure 15: Final electron phase space with the candidate-bunch selection highlighted

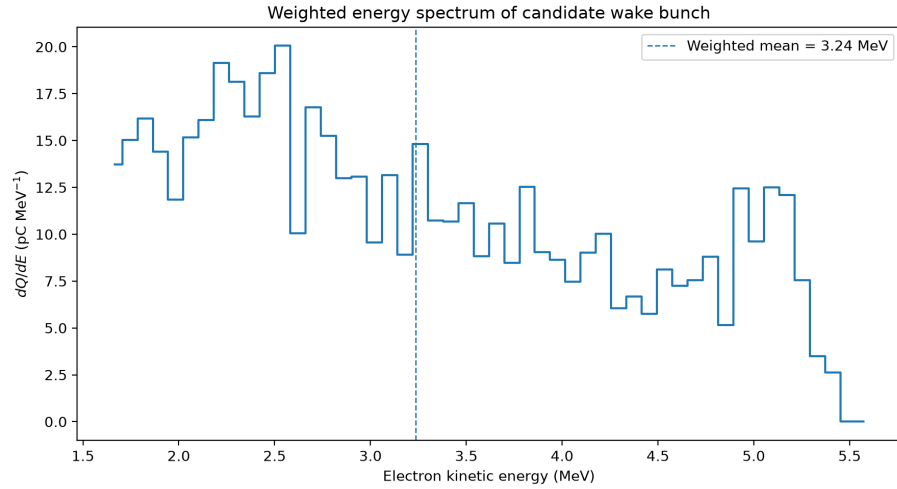


Figure 16: Weighted energy spectrum of the candidate wake bunch

Diagnostic 13.2: candidate-bunch selection sensitivity

The spatial and momentum limits were varied to test whether the candidate-bunch conclusion depended on a narrow choice of selection boundaries.

Spatial-window sensitivity

With $u_z \geq 4$ fixed:

ξ range	Q (pC)	Weighted mean energy (MeV)	Relative RMS spread	Macroparticles
-21 to -14 μ m	43.599	3.228	33.18%	2,626
-20 to -14 μ m	43.254	3.235	33.14%	2,609
-19 to -14 μ m	40.710	3.279	33.24%	2,510
-20 to -13 μ m	43.254	3.235	33.14%	2,609
-20 to -15 μ m	43.254	3.235	33.14%	2,609

The spatial cut is not especially sensitive. Moving the boundaries by about $1\mu\text{m}$ only changes the charge by a few pC, while the mean energy and relative spread hardly move. Shifting the right-hand edge does nothing at all in several cases, which shows that the selected high- u_z branch already ends before that boundary.

Momentum-cut sensitivity

With $-20 \leq \xi \leq -14\mu\text{m}$ fixed:

Minimum u_z	Q (pC)	Weighted mean energy (MeV)	Relative RMS spread	Macroparticles
3.5	50.433	3.009	37.94%	3,376
4.0	43.254	3.235	33.14%	2,609
4.5	37.208	3.449	28.99%	2,048
5.0	31.316	3.676	25.08%	1,594

The momentum cut matters more, which makes sense because it passes through the lower edge of the branch itself. Increasing the threshold removes more of

the lower-energy population, so the selected charge falls while the mean energy rises and the relative spread gets smaller.

At the final snapshot, all the tested momentum cuts give the same general picture: tens of pC, a few MeV and a broad spectrum. This does not mean that the charge is equally insensitive to the cut earlier in the run. A momentum cut-off of $u_z \geq 4$ was kept as the nominal choice because it follows the visible separation in phase space and is close to the 95th percentile of the full saved population. The exact bunch numbers should still be read as selection-dependent.

Associated file:

- `analysis/validation/test_bunch_selection_sensitivity.py`

Diagnostic 13.3: evolution of the candidate population

The final-snapshot selection was then applied to every saved iteration to see how the candidate population developed during the run. At each iteration, the measured laser position was used to define

$$\xi = z - z_{\text{laser}},$$

with saved electrons satisfying $-20 \leq \xi \leq -14 \mu\text{m}$ and $u_z \geq 4$ selected as candidates. The charge was calculated from the macroparticle weights, while the mean kinetic energy and relative RMS spread were also weighted. Iterations with no selected particles have zero charge; their mean energy and spread are undefined and are not plotted.

The selected charge stayed very small for most of the run, then grew rapidly towards the final snapshot. Three iterations show the main change:

Iteration	Selected macroparticles	Selected charge (pC)	Weighted mean energy (MeV)	Relative RMS spread
1450	144	0.00223	2.416	29.7%
1500	243	2.628	1.907	5.0%
1750	2,609	43.254	3.235	33.1%

The increase between iterations 1450 and 1500 requires additional analysis. The number of selected macroparticles rises from 144 to 243, but the charge rises much more sharply because the selected set at iteration 1500 contains particles with much greater macroparticle weights. Raising the momentum threshold shows where the charge at iteration 1500 lies:

Iteration	Charge for $u_z \geq 4$ (pC)	Charge for $u_z \geq 4.5$ (pC)	Charge for $u_z \geq 5$ (pC)
1450	0.00223	0.00142	0.00106
1500	2.628	0.231	0.00112
1750	43.254	37.208	31.316

At iteration 1500, approximately 2.397 pC, or 91% of the nominal selected charge, lies in $4 \leq u_z < 4.5$. The charge above $u_z = 5$ is almost unchanged from iteration 1450. The lower mean energy and spread at iteration 1500 therefore reflect the change in which particles dominate the weighted selection; they do not show that the electrons selected at iteration 1450 decelerated. By iteration 1750, the higher-momentum part of the branch is much more substantial: 31.316 pC remains when the threshold is raised to $u_z \geq 5$.

The charge evolution under all three momentum cuts adds to this comparison. Appreciable weighted charge first appears under the nominal selection around iteration 1500, initially concentrated close to $u_z = 4$. The charge above $u_z = 5$ remains very small at that snapshot, then grows rapidly in the following saved outputs. Alongside the phase-space location behind the laser, this provides stronger evidence that a candidate accelerated population develops in the wake during the run. It locates the change in the selected distribution, not the injection time of individual electrons. The three-cut plot uses a logarithmic charge axis and shows only snapshots with nonzero selected charge; the earlier zero-charge snapshots remain visible in the full bunch-evolution plot.

Selected charge was also analysed alongside the measured laser spot size and peak a_0 on a common propagation-distance axis. The laser expands and its peak a_0 falls while the selected charge grows late in the run. This establishes when the changes occur relative to one another, but does not show that the change in laser spot size caused the charge increase.

The selection is repeated independently at each snapshot, so these measurements are not trajectories of the same electrons. Particle IDs were not saved, and the particle diagnostic excluded electrons with $u_z < 1$. The figures also show longitudinal position and momentum without checking the radial distribution of the selected charge. At the final snapshot, the visible high-momentum branch extends across the sharp on-axis E_z sign change near $z \approx 86 \mu\text{m}$; the unweighted scatter plot cannot show how much selected charge lies on either side, and the on-axis field need not equal the field at each particle. The results are consistent with late capture and acceleration in the wake, but cannot establish precisely when individual electrons were injected, whether all selected electrons are currently in an accelerating phase, how long they remained trapped, or what caused the charge to increase.

Associated files:

- `analysis/electrons/analyse_bunch_evolution.py`

- analysis/validation/test_bunch_cut_sensitivity.py
- figures/electrons/bunch_evolution.png
- figures/electrons/laser_bunch_comparison.png
- figures/electrons/charge_evolution_cutoff_comparison.png
- figures/validation/check_momentum_histogram/candidate_momentum_histogram_1500.png

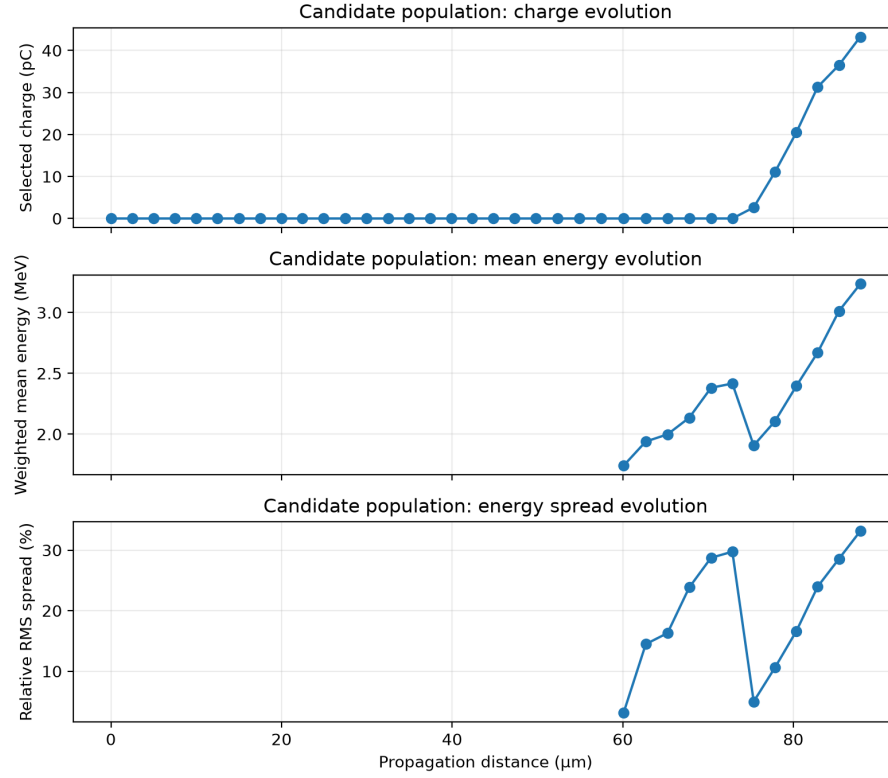


Figure 17: Charge, weighted mean energy and relative spread of the candidate population through the saved snapshots

Overall interpretation

The diagnostics now present a clear picture of this first run. At the final snapshot, the laser sits near $z \approx 103 \mu\text{m}$, with the clipped density plot clearly showing a depleted cavity and a dense sheath which closes near $z \approx 86.5 \mu\text{m}$. As this is almost exactly where the strongest negative on-axis E_z appears, these features together provide strong evidence that a wake formed.

The final phase-space plot also puts the high-momentum branch in this trailing region, and separates it from the regular oscillatory structure that moves with the laser. These are distinct phase-space structures, although the saved

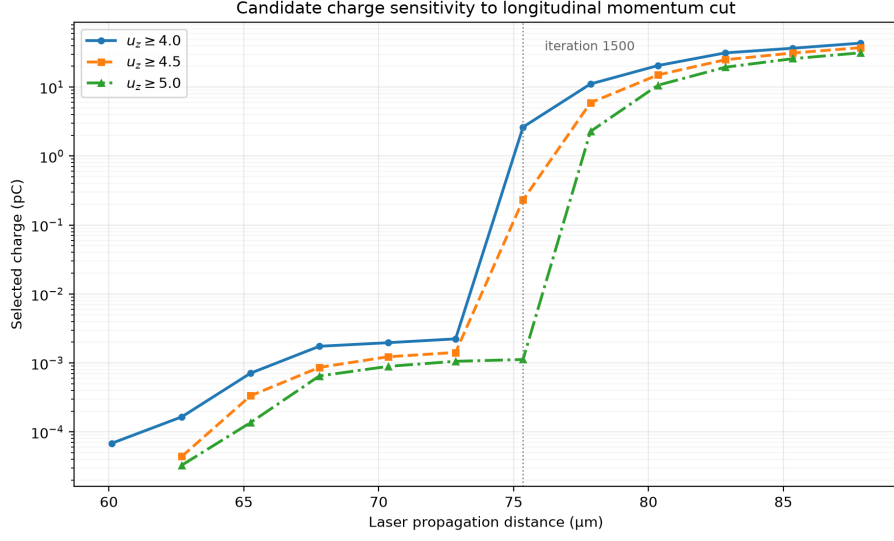


Figure 18: Selected charge under three momentum cuts during the late part of the run

snapshots cannot establish the histories of individual electrons.

The candidate selection remained at very low charge until late in the run. Appreciable selected charge first appears around iteration 1500 under the nominal $u_z \geq 4$ cut, although this initial rise is concentrated in $4 \leq u_z < 4.5$ and depends strongly on the threshold. Charge then grows under the stricter cuts too; by the final snapshot, 31.316 pC remains even with $u_z \geq 5$. Together with the trailing phase-space branch and the wake structure, this supports the emergence and growth of a candidate accelerated population in the wake. It is consistent with late capture and acceleration, but these independently selected snapshots do not identify an injection time or demonstrate continuous trapping for individual electrons. The radial distribution and the selected charge's position relative to the accelerating field remain to be checked.

One of the main lessons from the baseline is that neither peak E_z nor $E_{\max}$ tells the whole story. The wake can strengthen while the maximum saved-particle energy falls, and the full saved spectrum mixes laser-associated and wake-associated electrons. Bunch metrics only become useful after the relevant population has been identified in phase space.

For the nominal late-time bunch selection, the result is approximately

$$Q_{\text{bunch}} \approx 43 \text{ pC},$$

$$\bar{E}_{\text{bunch}} \approx 3.24 \text{ MeV},$$

with a relative RMS energy spread of approximately

$$\frac{\sigma_E}{\bar{E}} \approx 33$$

The exact numbers move with the lower momentum cut, but not enough to change the main conclusion: the late wake-associated population is at the tens-of-pC level, has a characteristic energy of a few MeV and is broad-spectrum rather than cleanly quasi-monoenergetic.

Limitations and next checks

- The initial $16.2 \mu\text{m}$ peak-separation result is not an unambiguous wake-wavelength measurement because the second detected E_z maximum lies within the laser region.
- The laser-propagation diagnostic also explains the reduction in the transverse-field envelope identified earlier. The spot size increased by about 29.5% while the peak a_0 fell by about 21.8%, with $a_0 w$ remaining approximately constant. This is mainly consistent with diffraction-driven transverse expansion rather than direct evidence of pump depletion. The propagation remained close to the vacuum prediction, although the short distance travelled through the full-density plateau prevents a strong conclusion about longer-term self-focusing or matched guiding.
- The exact $127.6 n_0$ density spike and -680 GV m^{-1} field minimum are highly localised. Their associated large-scale structures are physically coherent and the E_z spike survives modest spatial averaging, but their exact amplitudes are not yet numerically converged.
- The z - u_z scatter plots are unweighted visualisations. Macroparticle weights are included in charge and energy distributions but not in the displayed scatter-point density.
- The particle diagnostic excludes electrons with $u_z < 1$, so none of the saved-particle plots represent the complete plasma-electron population.
- The candidate bunch is defined using explicit spatial and momentum cuts. The spatial selection is relatively insensitive to small changes, while the inferred charge, mean energy and spread vary more noticeably with the lower momentum threshold. The selected charge has not yet been checked against radial position or against the local accelerating-field phase.
- The laser-propagation diagnostic measures spot size and peak a_0 from the transverse fluence and field envelope. It does not directly measure total pulse energy or pump depletion.
- Numerical convergence with respect to longitudinal/radial resolution, macroparticle sampling and retained azimuthal modes has not yet been

tested for either the sharp field and density peaks or the selected bunch charge.

- Before parameter optimisation, at least a small numerical-validation comparison should be performed so that optimisation metrics are not dominated by resolution, sampling or mode choices.
- Individual particle IDs were not tracked between saved snapshots, so the phase-space evolution distinguishes changing populations by their structure and position but does not provide individual electron trajectories.